AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2011

Biology Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL NINE questions. Questions 7, 8 and 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. What will happen if the walls of the blood vessels become thick due to smoking? (2 Marks)

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b. A chef is cooking in a kitchen in an extremely hot environment. (2 Marks)

Give TWO changes that will take place in his body when he carries the dishes to the air-conditioned dining room.

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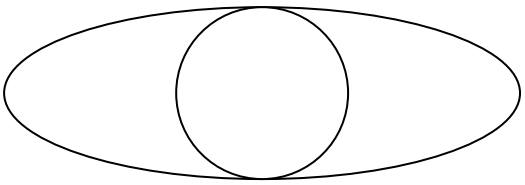
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Q.2. (Total 4 Marks)

a. A person from a dark room comes out into sunlight. (2 Marks)

Draw the state of longitudinal muscles and size of pupil on the given sketch to show the control of bright light entering into the eye.



b. A person is unable to see clearly in dim light. State the disease he is suffering from along with its cause. (2 Marks)

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Q.3. (Total 5 Marks)

a. The presence of discs between individual vertebral bones is a unique feature of the backbone. Why is the backbone, in this aspect, different from other bones of the body? (2 Marks)

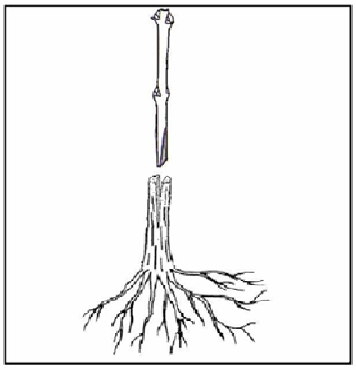
1. _____

_____

2. _____

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b. The diagram shows the process of grafting in plants. (3 Marks)



i. Identify the scion in the diagram.

ii. What care should be taken when scion and stock are tied together?

1. _____

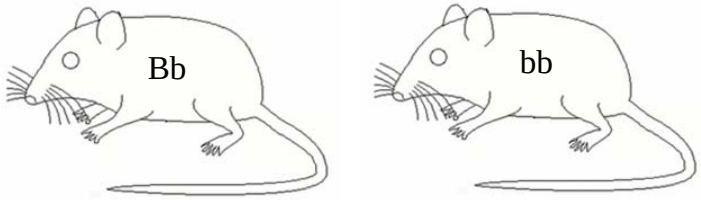
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2. _____

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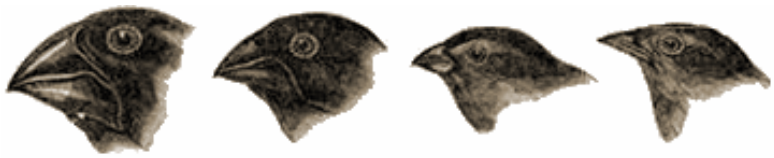
Q.4. (Total 4 Marks)

a. The mice shown below are crossed. (2 Marks)



What will be the expected genotype of the off spring in F1 generation?
(You are required to show the working of the cross)

b. Charles Darwin identified 13 species of finches (birds) in the Galapagos Islands. (2 Marks)



Which phenomenon is responsible for the occurrence of such variety in finches?

What could be the possible reason for the change in the size and shape of the beak?

Q.5. (Total 4 Marks)

a. A crop is sprayed with an inorganic insecticide. How will this insecticide enter into a food chain and reach the consumer? (3 Marks)

b. Name any TWO gases that prevent the emission of heat waves into space. (1 Mark)

Q.6. (Total 4 Marks)

a. What are single-cell proteins and from where are they obtained? (2 Marks)

b. Name any TWO plants which are commonly used for obtaining drugs. (2 Marks)

Q.7. (Total 5 Marks)

EITHER

a. Explain which structural features of alveoli help them to perform gaseous exchange efficiently.

OR

b. What actions will be taken by the kidneys and associated glands of the body to keep the osmotic pressure constant:

i. when loss of water occurs through sweating.
ii. after drinking large amounts of water.

Q.8. (Total 5 Marks)

EITHER

a. An elderly woman with poor calcium intake in her diet ultimately develops bone related defects.

Describe TWO defects of the skeletal system the woman is most likely to be suffering from. Also mention TWO ways to prevent such a condition.

OR

b. Explain what conditions should be kept ideal to allow plants to experience uninterrupted growth.

Q.9. (Total 5 Marks)

EITHER

a. One of the aims of genetic engineering is to produce crops that have improved agricultural or food quality traits.

Discuss the significance of genetic engineering with at least FOUR examples.

OR

b. Alcohol is considered as a socially unacceptable drug. Discuss this statement with reference to short-term and long-term effects of alcohol abuse in humans.

END OF PAPER